YEAR 10 PHYSICS, AQA GCSE, LESSON 7 OF 12

ELECTRIC CIRCUITS

LESSON 07 RESISTORS IN SERIES AND PARALLEL

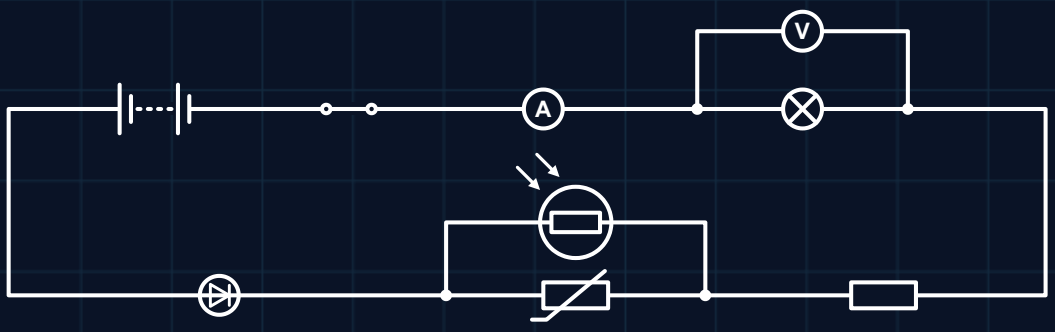
REQUIRED PRACTICAL

Two identical resistors, three ways to connect them. Which arrangement lets the most current flow?

IN THIS LESSON YOU WILL

- measure the total resistance of resistors in series and in parallel
- compare results with $R_{total} = R_1 + R_2$
- explain the results using paths and current
- evaluate the measurements

ANSWER EDITION FOR TEACHERS AND SELF-MARKING



NAME _____ CLASS _____

TEACHER _____ TARGET GRADE _____

START HERE

HOW THIS BOOKLET WORKS

This is the booklet for **Lesson 7** of the Electric Circuits unit. It builds on the lessons before it; the "Do now" questions and the quick answers at the back help you check what you remember. Keep it: it is your classwork book *and* your revision guide.

DO NOW
Quick questions from earlier lessons. Remembering again is what makes it stick.

LEARN IT
Explanations and diagrams. Fill in the slide examples with your teacher, then try "Your turn" alone.

TRY IT
Practice that builds up: first with support, then on your own. Quick checks from the slides follow some lessons.

LAB
Record measurements from a virtual lab or a required practical.

EXAM CORNER
An AQA-style question with a tip on how the marks are given.

REVISE IT
Summary, cover-and-answer questions and an "I can" check to colour in.

! **Revising at home?** (1) Read the Learn it pages. (2) Cover them and answer the "Cover the page" questions in Revise it. (3) Check with the quick answers at the back.

THE ROUTE THROUGH THE UNIT (THIS LESSON IS HIGHLIGHTED)

01 Complete circuits and symbols

02 Current and charge

03 Potential difference

04 Resistance and V = I R

05 Resistance of a wire RP

06 Series and parallel circuits

07 Resistors in series and parallel RP
Required practical: measure total resistance of two resistors in series and in parallel

08 I–V graphs

09 Investigating I–V characteristics RP

10 Thermistors, LDRs and sensors

11 Power and energy transfer

12 Mains, safety and the National Grid

Lesson

07

RESISTORS IN SERIES AND PARALLEL

AQA Combined RP 15; Physics RP 3; Part B

BIG QUESTION **Two identical resistors, three ways to connect them. Which arrangement lets the most current flow?**

BY THE END YOU CAN

- measure the total resistance of resistors in series and in parallel
- compare results with $R_{total} = R_1 + R_2$
- explain the results using paths and current
- evaluate the measurements

total resistance **prediction** **uncertainty** **resolution** **evaluate**

RETRIEVE

DO NOW

From earlier lessons. No notes!

1

Total resistance of 10 Ω and 15 Ω in series?

25 Ω

2

In parallel, what is the same for each branch?

The p.d.

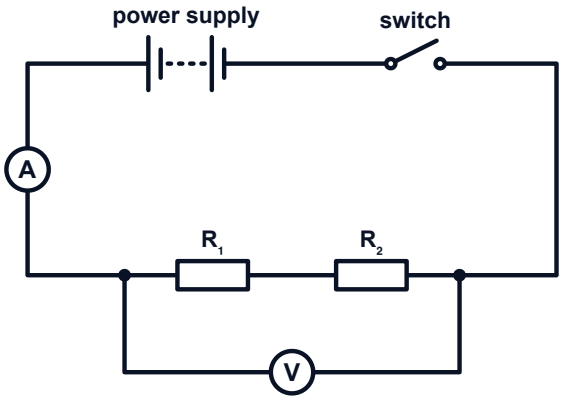
3

The supply current is 0.9 A. One branch carries 0.4 A. What does the other carry?

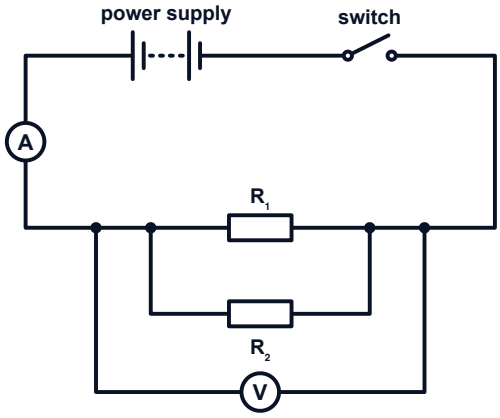
0.5 A

EXPLAIN

PLAN THE PRACTICAL



Series: the voltmeter goes across both resistors.



Parallel: the voltmeter goes across the pair.

Measure the p.d. **across the whole combination** and the current **from the supply**. Then the total resistance is **R = V ÷ I**.

1. Connect R_1 on its own. Measure V and I. Repeat for R_2 on its own.
2. Connect R_1 and R_2 in series. Measure V across both, and I.
3. Connect R_1 and R_2 in parallel. Measure V across the pair, and the supply current.
4. Calculate R = V ÷ I for each arrangement.

STAY SAFE

- Resistors can get hot. Switch off between readings.
- Keep the p.d. low, and check the circuit before switching on.

1. **Predict** before you measure. Rank R_1 alone, series and parallel from the *largest* to the *smallest* total resistance.

series > one resistor > parallel

RESULTS

Answer edition: example results with two 10 Ω resistors at 3.0 V.

6 Record your readings.

Arrangement	p.d. / V	current / A	R = V ÷ I / Ω
R_1 on its own	3.0	0.30	10
R_2 on its own	3.0	0.30	10
R_1 and R_2 in series	3.0	0.15	20
R_1 and R_2 in parallel	3.0	0.60	5.0

ANALYSE AND EVALUATE

2 Compare your series result with $R_1 + R_2$. Do they agree?

Series ≈ 20 Ω and $R_1 + R_2$ = 10 + 10 = 20 Ω, so they agree (within measurement error).

3 Explain why the total resistance in parallel is *less* than either resistor on its own.

Adding a second branch gives the charge …

… another path. At the same p.d. the supply current increases (0.60 A instead of 0.30 A).

R = V ÷ I, so the total resistance is smaller.

4 Suggest one source of error in the measurements and one way to reduce it.

e.g. resistance of the leads and connections, or the resolution of the meters, or the resistors warming up. Use tight, short connections, meters with better resolution, and switch off between readings.

5 **EXTRA PRACTICE** Resistors of 5.0 Ω, 10 Ω and 15 Ω are connected in series with a 6.0 V supply. Calculate the current.

TOTAL R	**5.0 + 10 + 15 = 30 Ω**
EQUATION	**I = V ÷ R**
SUBSTITUTE	**I = 6.0 ÷ 30**
ANSWER	**0.20 A** unit:

CHECKPOINT

QUICK CHECK

Slide 31, Mission 5. Answer each question, then check it as your teacher reveals the answer.

1
5 Ω and 15 Ω are in series. Calculate the total resistance.

5 + 15 = 20 Ω

2
In series with a 12 V supply, one part has 5 V across it. What does the other part have?

7 V

3
Two branches carry 0.4 A and 0.6 A. What is the supply current?

0.4 + 0.6 = 1.0 A

4
What is the p.d. across each parallel branch?

The full supply p.d.

5
A resistor is added in parallel. What happens to the total resistance?

It decreases.

6
6.0 V supply, 0.20 A, R_1 = 12 Ω. Calculate R_2 in series.

R_{total} = 6.0 ÷ 0.20 = 30 Ω;

30 − 12 = 18 Ω

AQA-STYLE

EXAM CORNER

Write in full sentences. Show every step of a calculation.

Get the marks: 'Explain' needs a chain: another path → more current at the same p.d. → lower total resistance.

7 Two identical resistors each have a resistance of 12 Ω. (a) Calculate their total resistance in series. (b) A student measures 6.0 Ω when they are connected in parallel. Explain why this is less than 12 Ω. (c) State one variable the student should keep the same for a fair comparison. **[5 marks]**

(a) 12 + 12 = 24 Ω (1). (b) Parallel gives another path for the charge (1), so at the same p.d. the total current is larger (1), and R = V ÷ I is smaller (1). (c) The supply p.d., the same resistors, or the temperature (1).

REVIEW LATER

REVISE IT

Use this box after the lesson and before tests.

SUMMARY

- Find total resistance from the **supply current** and the **p.d. across the combination**: R = V ÷ I.
- Series: the total is the sum (R_1 + R_2): bigger than either.
- Parallel: the total is smaller than either resistor.
- Evaluate by naming a specific source of error and a specific improvement.

COVER THE PAGE AND ANSWER

1 How do you find the total resistance of a combination in the lab?

2 Two 10 Ω resistors in series?

3 Is two 10 Ω in parallel more or less than 10 Ω?

4 Give one source of error in this practical.

I CAN…

- set up series and parallel combinations with the meters in the right places
- calculate total resistance from V and I
- explain why series adds and parallel lowers resistance
- suggest a specific improvement to a practical

REFERENCE

EQUATION TOOLKIT

The equations you have met so far. You will meet these one at a time. Learn each equation **in words** as well as symbols: AQA gives you an equation sheet in the exam (it does not include $R_{total} = R_1 + R_2$), but knowing the equations by heart makes every question faster.

$Q = I\,t$

Lesson 2

charge flow = current × time

C, A, s

Rearranged: I = Q ÷ t and t = Q ÷ I

$E = Q\,V$

Lesson 3

energy transferred = charge × p.d.

J, C, V

Rearranged: V = E ÷ Q and Q = E ÷ V

$V = I\,R$

Lesson 4

p.d. = current × resistance

V, A, Ω

Rearranged: I = V ÷ R and R = V ÷ I

$R_{total} = R_1 + R_2$

Lesson 6

total resistance in series

Ω

Rearranged: (series only)

FOUR STEPS FOR EVERY CALCULATION

1
Write down what you know, with units. Convert units first (minutes → seconds, kW → W, mA → A).

2
Write the equation you will use, in symbols.

3
Substitute the numbers, then rearrange if needed.

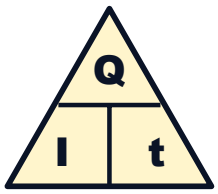
4
Give the answer with a unit. Ask: is it sensible?

CONVERT	HOW	EXAMPLE
minutes → seconds	× 60	3.0 min = 180 s
hours → seconds	× 3600	2.0 h = 7200 s
milliamps (mA) → amps (A)	÷ 1000	250 mA = 0.25 A
kilowatts (kW) → watts (W); kilohms (kΩ) → ohms (Ω)	× 1000	2.2 kW = 2200 W
kilovolts (kV) → volts (V)	× 1000	400 kV = 400 000 V

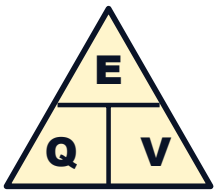
EQUATION TRIANGLES: HELP WITH REARRANGING

1. Cover the quantity you want to find.
2. The two left **side by side**? **Multiply** them.
3. One **above** the other? **Divide** the top by the bottom.

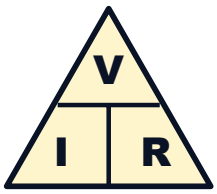
Example: to find I from Q = I t, cover I. Q is above t, so I = Q ÷ t. Triangles only help with rearranging: you still need to learn each equation.



Q = I t
Q = I × t
I = Q ÷ t
t = Q ÷ I



E = Q V
E = Q × V
Q = E ÷ V
V = E ÷ Q



V = I R
V = I × R
I = V ÷ R
R = V ÷ I

CHECK YOURSELF

QUICK ANSWERS

Answers to the “Do now” and “Cover the page” questions, so you can check yourself when revising. Answers to the longer questions are in your teacher's answer edition.

DO NOW

1. 25 Ω
2. The p.d.
3. 0.5 A

COVER THE PAGE

1. R = V ÷ I, with V across the combination and I from the supply.
2. 20 Ω
3. Less (it is 5 Ω).
4. Resistance of leads; meter resolution; heating.